
Digital Signal Processing

Lecture 1 Introduction

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DSP is Everywhere

- Sound applications
 - Compression, special effects, synthesis, recognition, echo cancellation,...
 - Cell Phones, MP3, Movies, Text-to-speech,...
- Communication
 - Modulation, coding, detection, equalization, echo cancellation,...
 - Cell Phones, dial-up modem, DSL modem, Satellite Receiver,...
- Automotive
 - ABS, Active Noise Cancellation, Cruise Control,

DSP is Everywhere

- Medical
 - Magnetic Resonance, Tomography, Electrocardiogram,...
- Military
 - Radar, Sonar, Space photographs, remote sensing,...
- Image and Video Applications
 - DVD, JPEG, Movie special effects, video conferencing,...
- Mechanical
 - Motor control, process control, oil and mineral prospecting,...

Limitations of Analog Signal Processing

- Accuracy limitations due to
 - Component tolerances
 - Undesired nonlinearities
- Limited repeatability due to
 - Tolerances
 - Changes in environmental conditions
 - Temperature
 - Vibration
- Sensitivity to electrical noise

Limitations of Analog Signal Processing

- Limited dynamic range for voltage and currents
- Inflexibility to changes
- Difficulty of implementing certain operations
 - Nonlinear operations
 - Time-varying operations
- Difficulty of storing information

Microprocessor

- Any CPU that is contained on a single chip
- Little chip is the heart of a computer. Often referred to as just the “processor”
- Does all the computations like adding, subtracting, multiplying, and dividing
- In PCs, most popular Intel Pentium chip
- In Macs, the PowerPC chip (Motorola, IBM, and Apple)

DSP, RISC, CISC Processor

- A processor is frequently categorized based on the width of its busses (4,8,16,32,64...)
- Clock Rate (i.e. at what rate does the processor execute instructions)
- Complexity of Instruction Set
 - CISC : Complex Instruction Set Computer
 - RISC : Reduced Instruction Set Computer

Digital Signal Processor

- A DSP is a “general purpose” processor with features specifically designed to make Signal processing applications fast and efficient

Design Options for Digital Systems

- **Special Purpose Hardware**
 - Custom ICs / ASICs
- **Software Programmable Processor**
 - Pentium, PowerPC, etc
- **FPGA (possibly with embedded general purpose microprocessor)**
 - Xilinx, Altera, etc
- **DSP**
 - TI, ADSP, etc

Comparison of Options

Specific HW

Gen Purpose HW

- NRE/Dev Cost
- Speed
- Flexibility
- Time to Market
- Production Cost



Embedded SW Design Flow

- Develop Code for a Target processor
- Since target is minimal (not much memory, I/O...etc. Code development done on a separate machine. (e.g a PC)
 - Cross Compiler / Assembler
 - Simulator
- Code then run in the target system and observed. Debug support programmed into the software

Emulation / Debugging

- In-Circuit Emulator
- Debug Kernel BIOS
- JTAG Emulation
- Interactively Run Code
 - Breakpoints
 - Single Step
 - Watch Variables
- Observe interaction with rest of system
- *Development environment is frequently processor specific*

Embedded Systems Characteristics

- **Real-Time**

Real, defined timing requirements for particular actions to be accomplished

- **Event Driven**

Actions of the system are in response to events, not a predefined sequence.

- **Resource constrained**

Memory Size, speed, power constrained

- **Special purpose**

Device must only perform certain well defined tasks

Embedded System Example

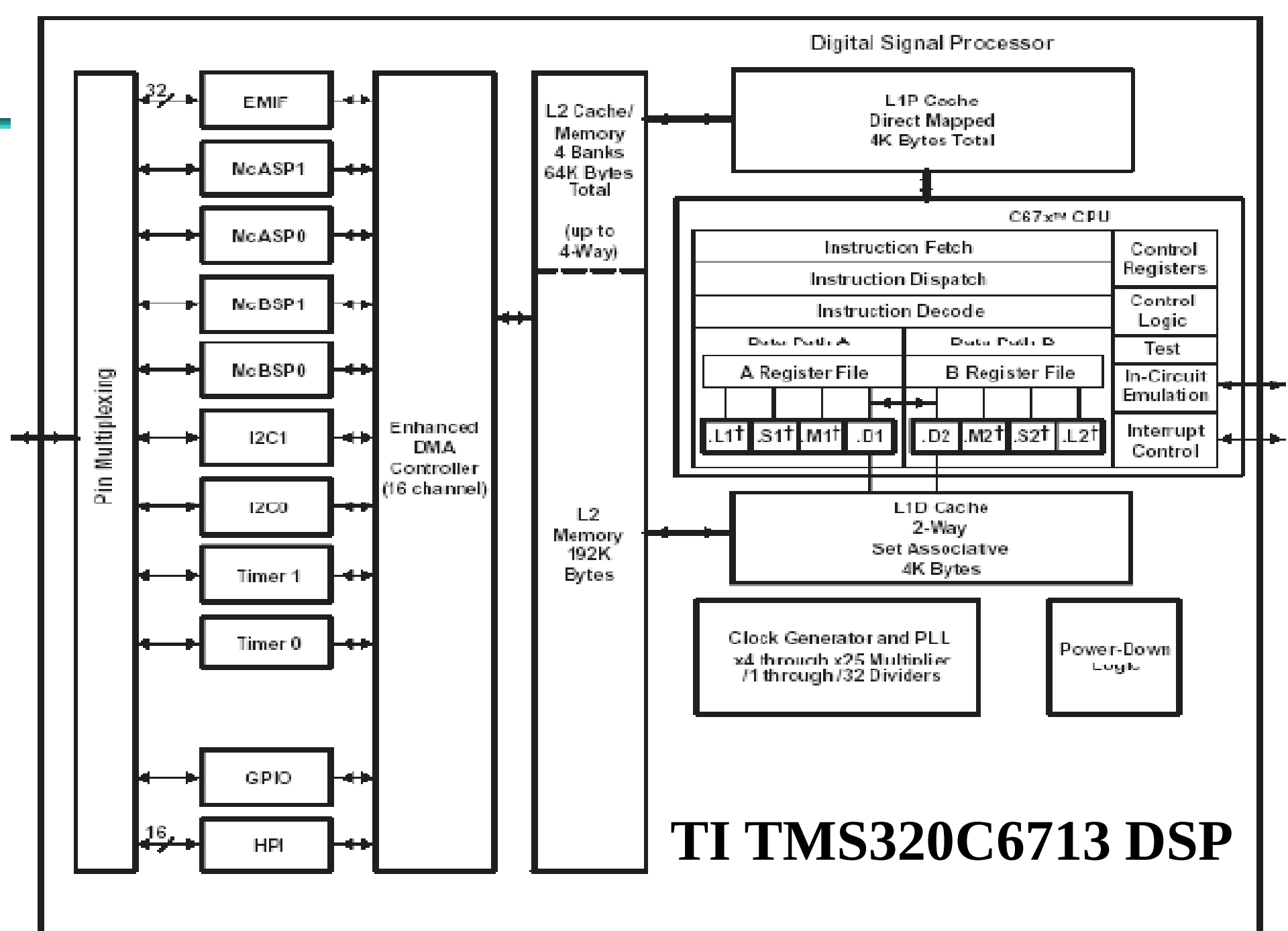
- Events :
 - Button Press
 - Knob Turned
 - New Sample needed by D/A converter
 - Data block available from CD drive



Denon Home Theater In a Box
Analog Devices SHARC DSP

The TMS320C6713 DSK





TI TMS320C6713 DSP Features

- EDMA Controller
- McBSP Serial Ports (I/O)
- Multiple Computation Units (8)
- Cache
- On-chip PLL
- Host Port Interface
- Timers
- Floating Point Units

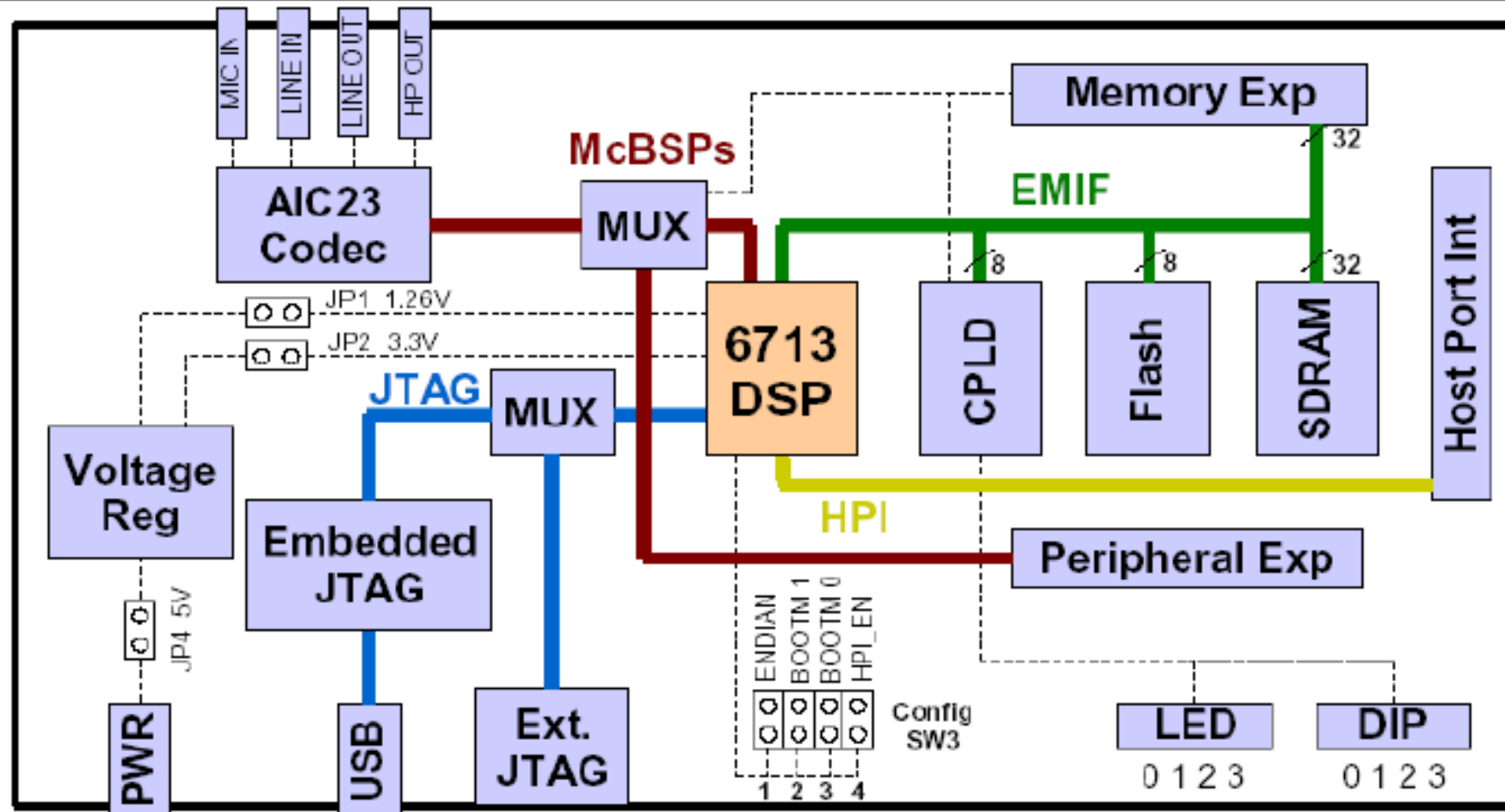
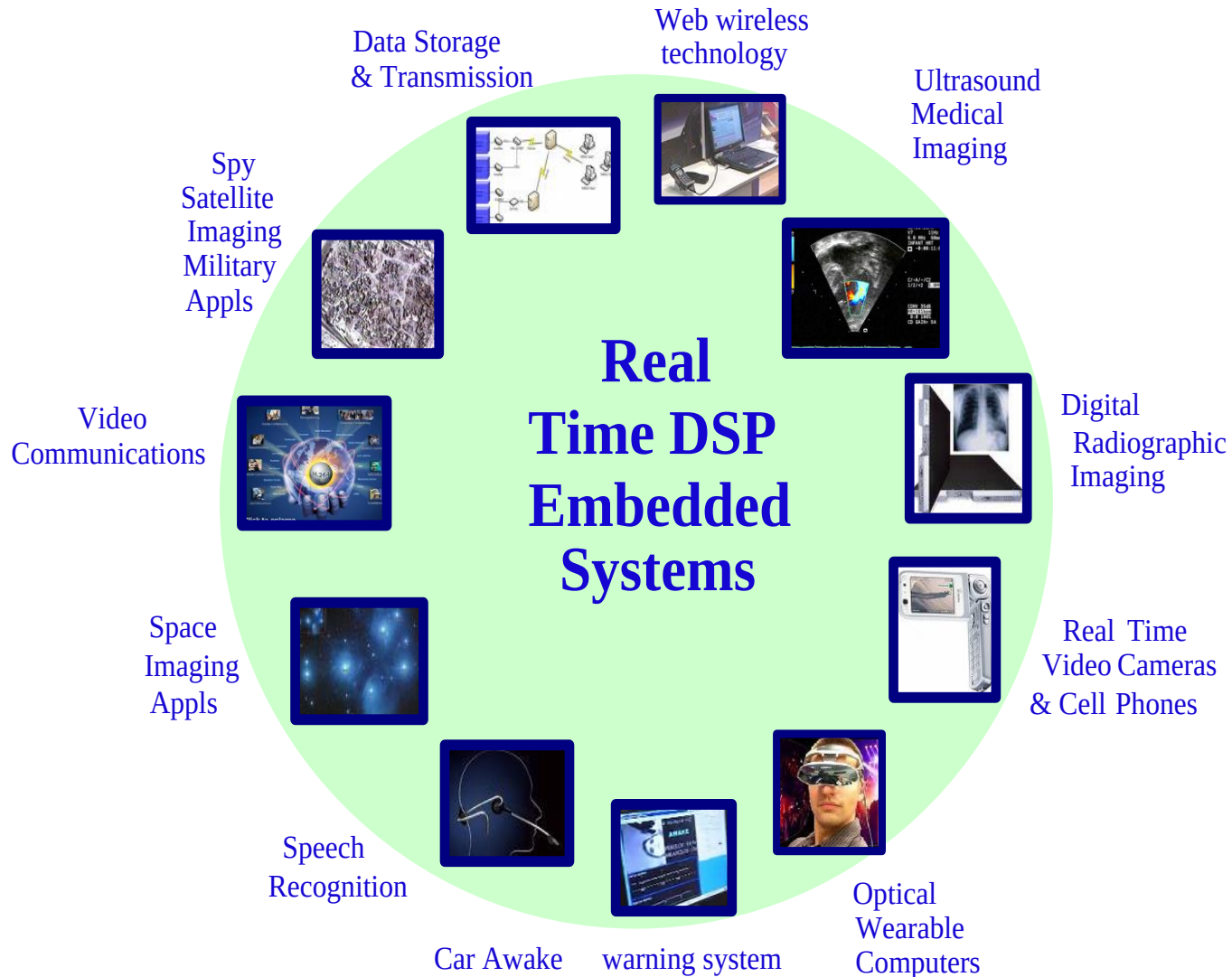
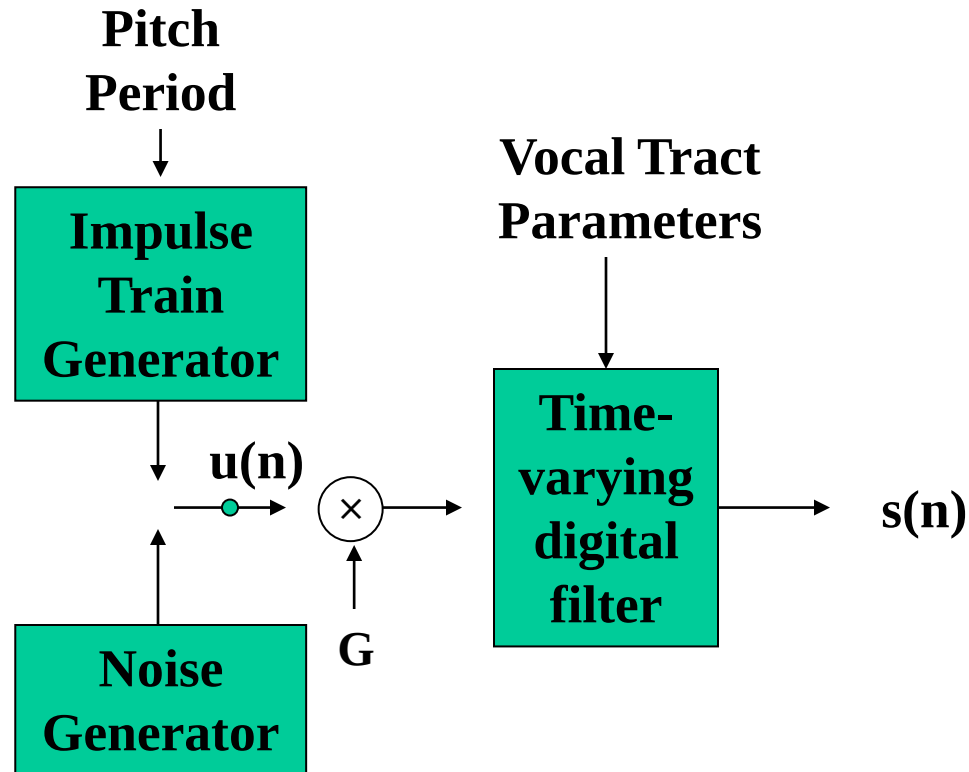


Figure 1-1, Block Diagram C6713 DSK

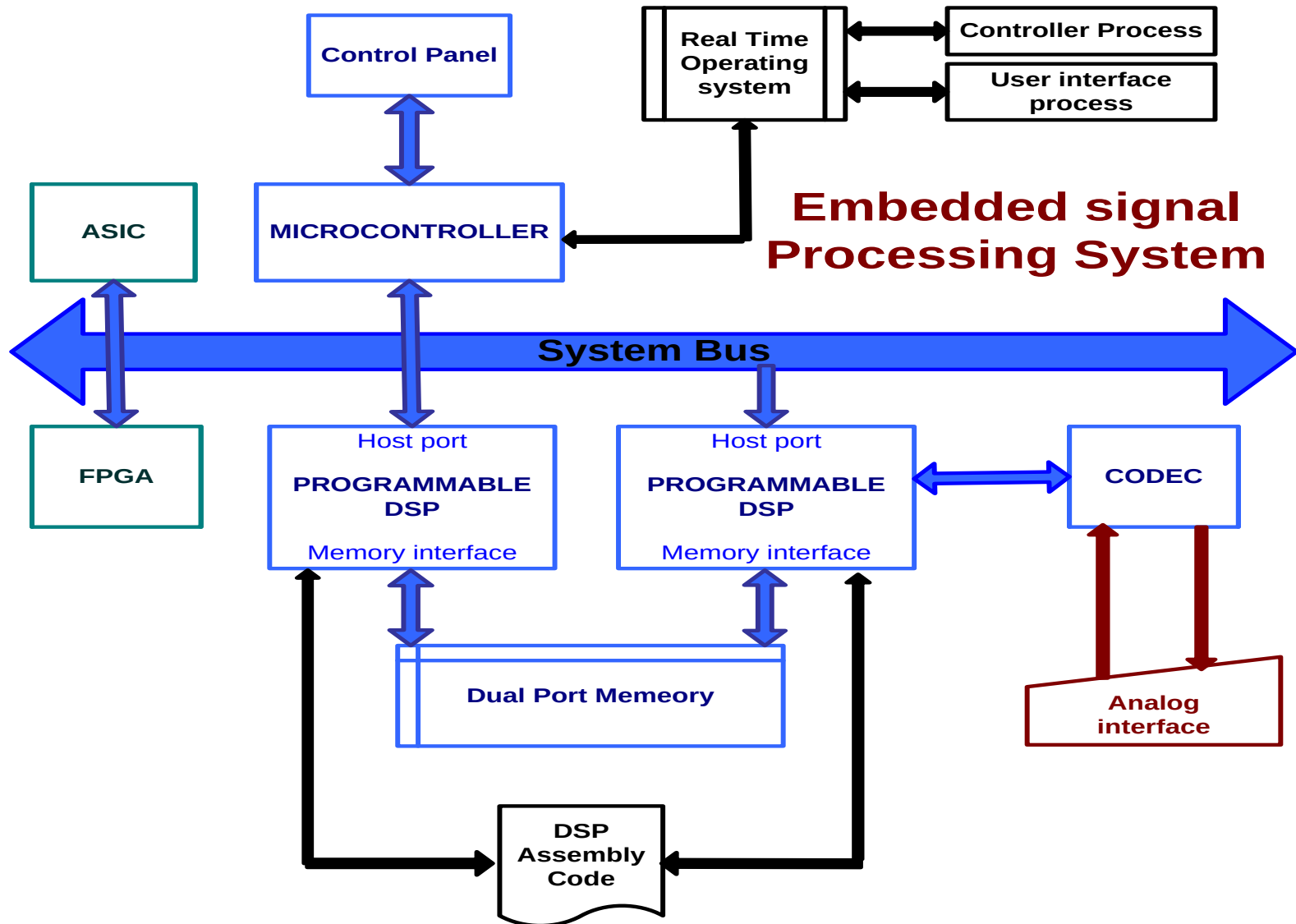
Typical DSP Applications



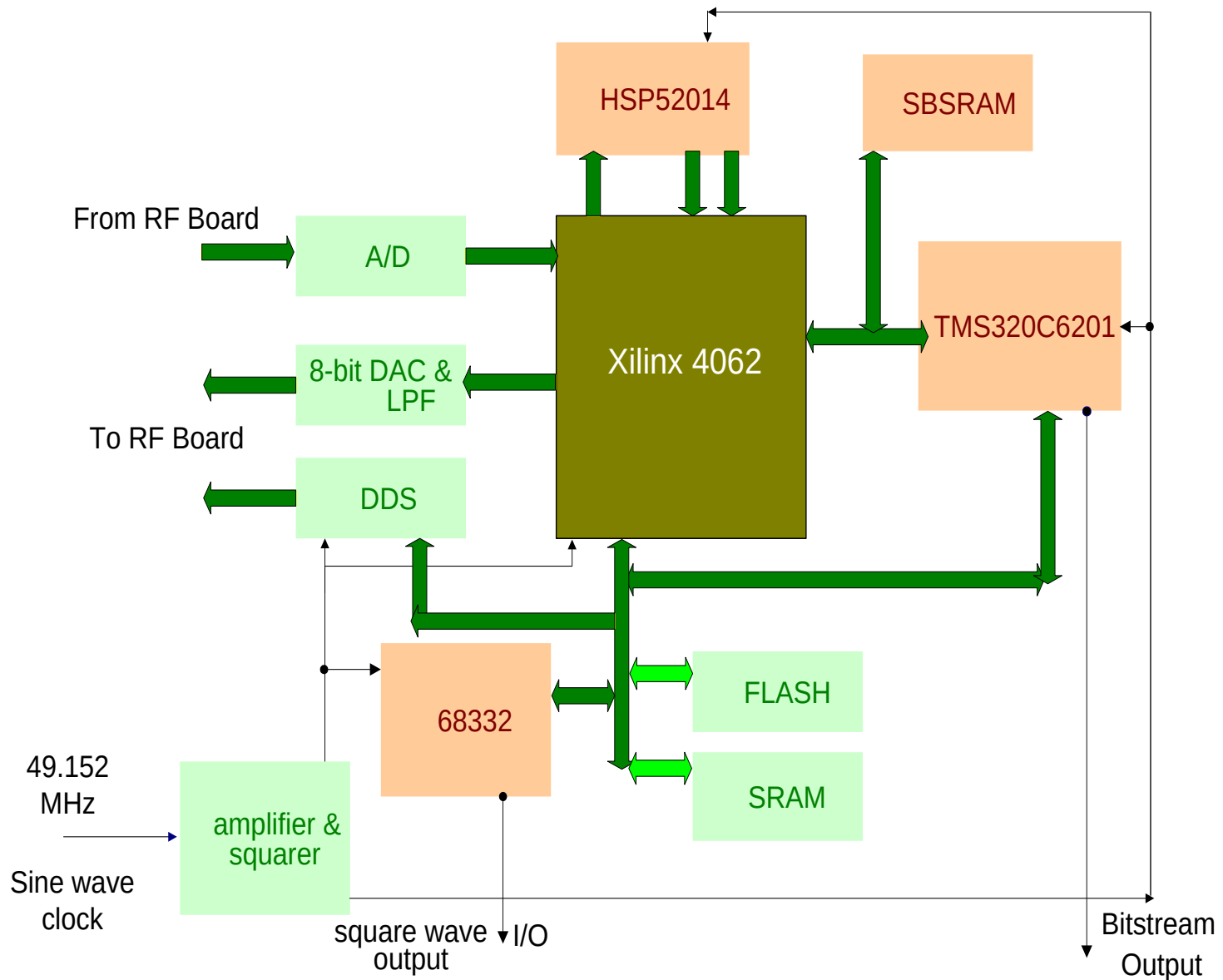
Example: Speech Modeling



An Embedded System

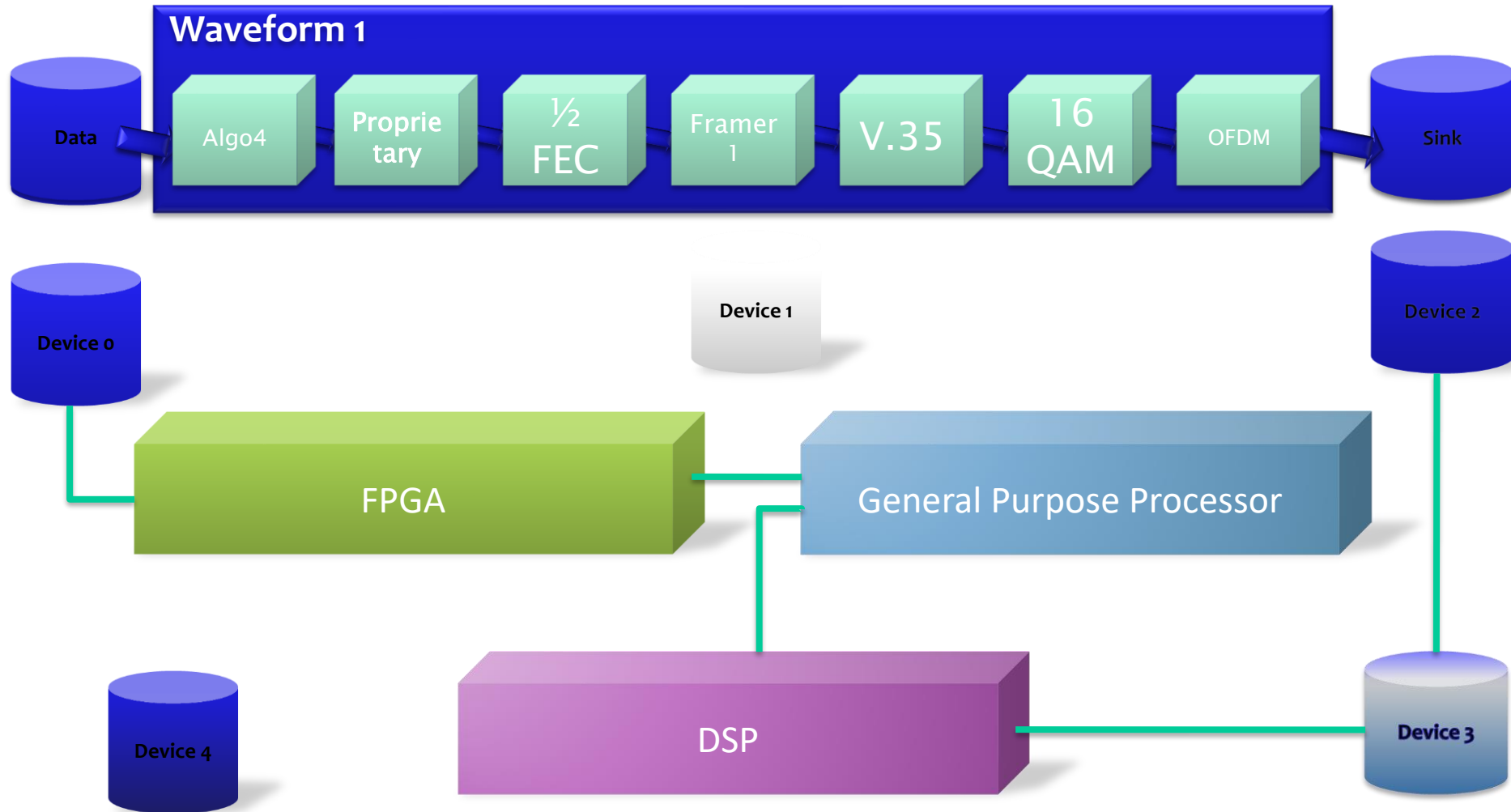


Example Embedded System

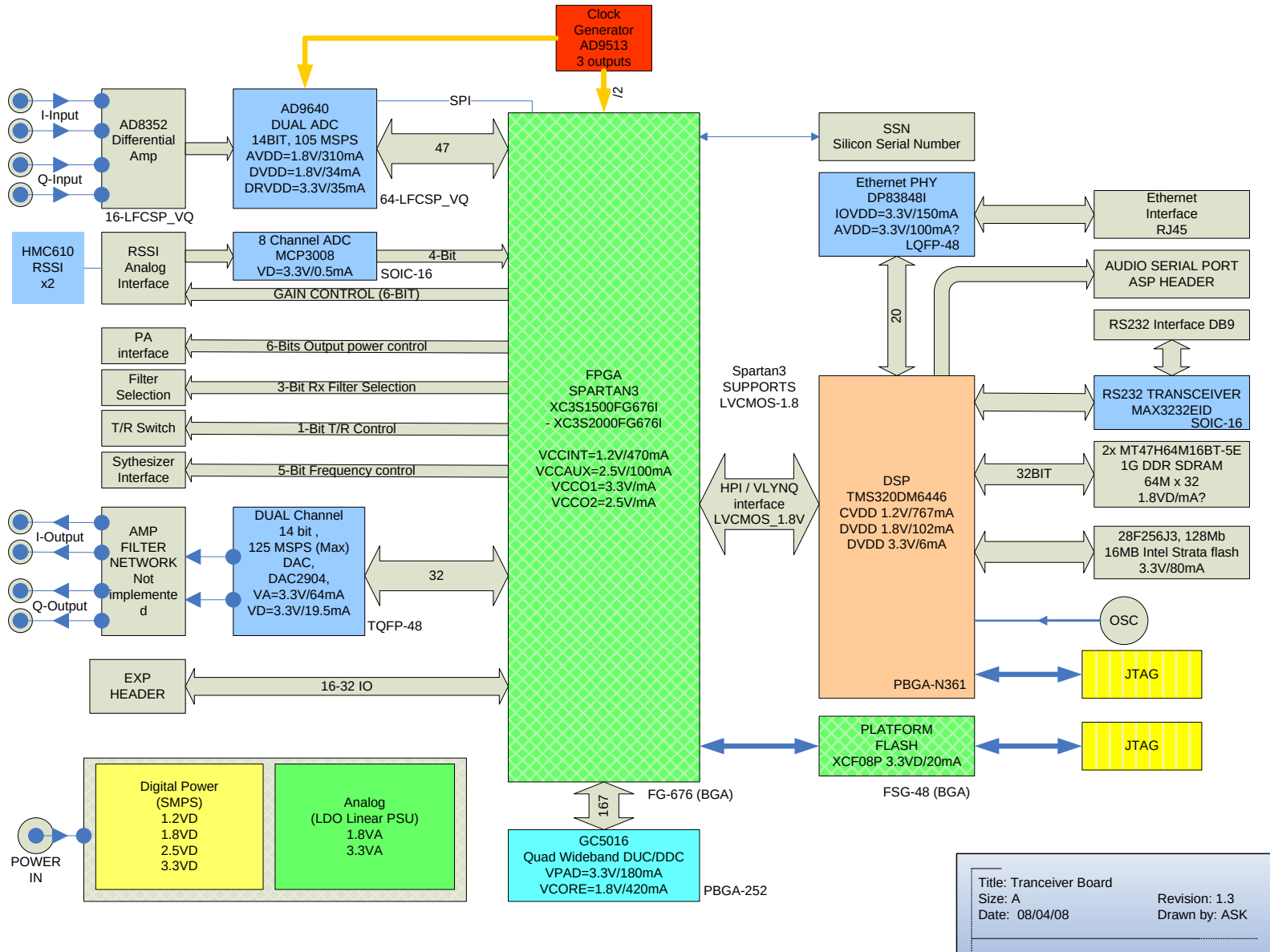


Software Defined Radio (SDR)

All configurable HW



SDR Board Design



Digital Signal Processing



- Analog input – analog output
 - Digital recording of music
- Analog input – digital output
 - Touch tone phone dialing
- Digital input – analog output
 - Text to speech
- Digital input – digital output
 - Compression of a file on computer

Pros of Digital Signal Processing

- Accuracy can be controlled by choosing word length
- Repeatable
- Sensitivity to electrical noise is minimal
- Dynamic range can be controlled using floating point numbers
- Flexibility can be achieved with software implementations
- Non-linear and time-varying operations are easier to implement

Pros of Digital Signal Processing

- Digital storage is cheap
- Digital information can be encrypted for security
- Price/performance and reduced time-to-market

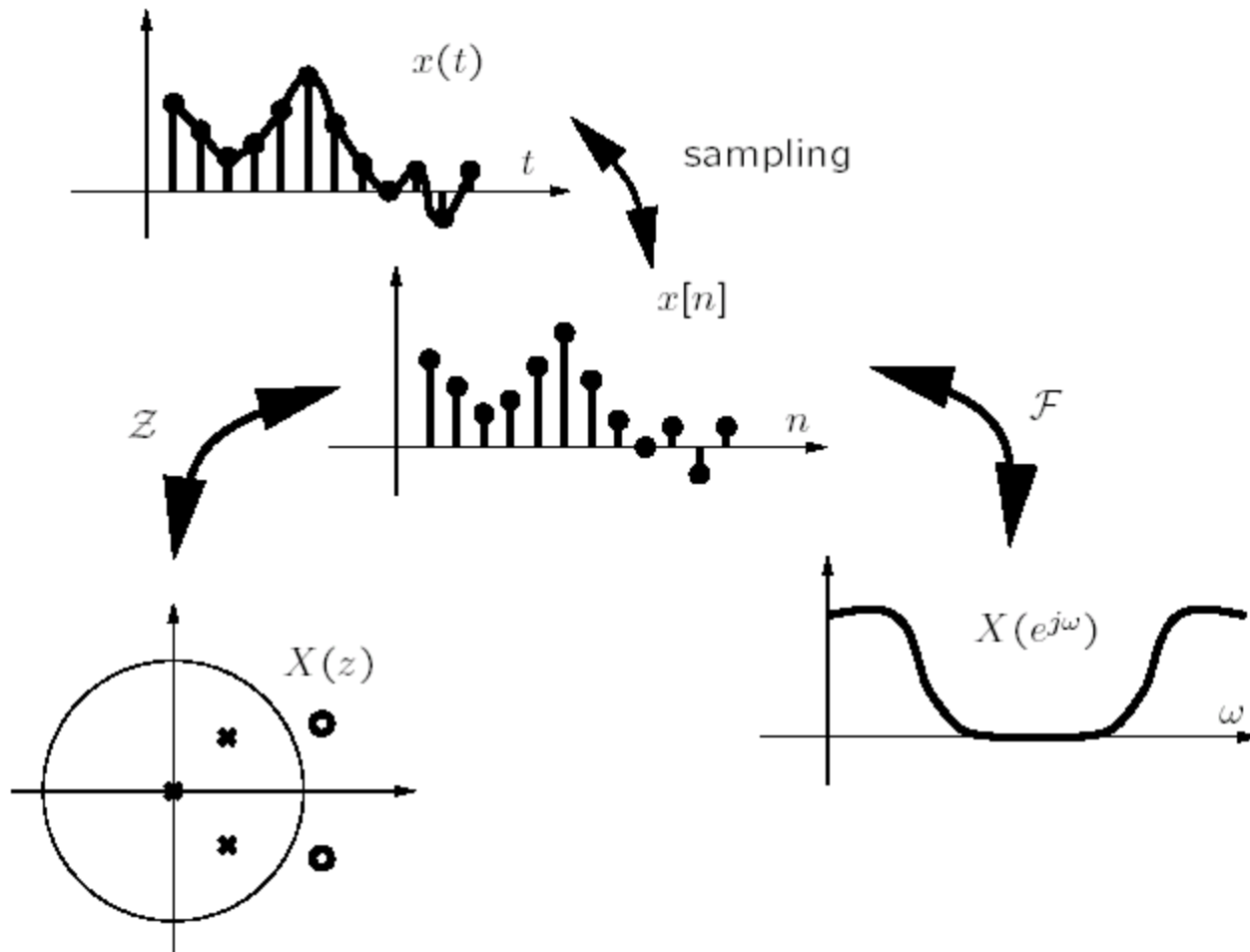
Cons of Digital Signal Processing

- Sampling causes loss of information
- A/D and D/A requires mixed-signal hardware
- Limited speed of processors
- Quantization and round-off errors

Course Objectives

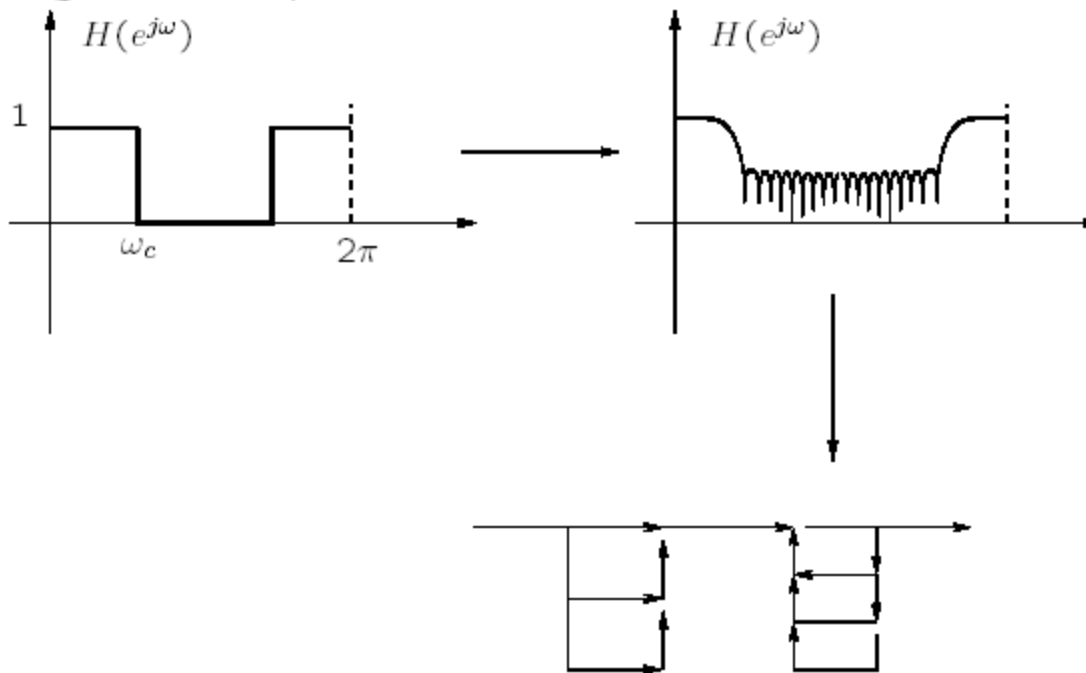
- To establish the idea of using computing techniques to alter the properties of a signal for desired effects, via understanding of
 - Fundamentals of discrete-time, linear, shift-invariant signals and systems in
 - **Representation and Analysis:** sampling, quantization, Fourier and z-transform;
 - **Implementation:** filtering and transform techniques;
 - **System Design:** filter & processing algorithm design

Course Outline



Course Outline

- DSP: A/D, D/A
- system design and implementation



- practical computation of $\mathcal{F}$: sampled version of the DTFT (DFT) and efficient computation.

Prerequisite

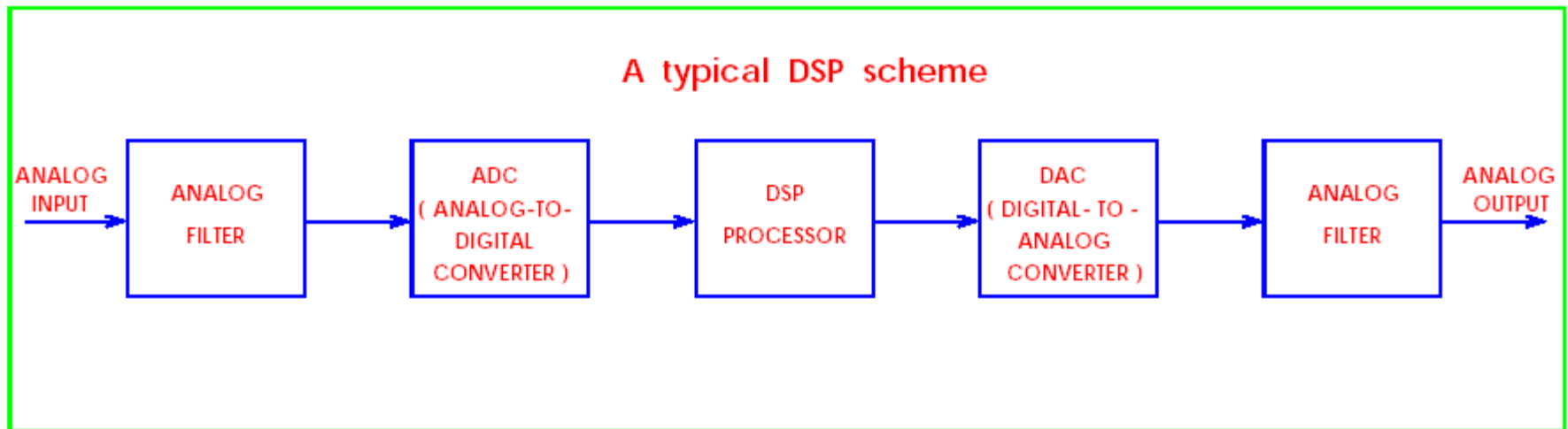
- A fundamental course in signal and system
 - Linear System analysis and transform analysis
 - convolution and filtering
 - Fourier transforms
 - Laplace and z transforms

Textbooks

- Oppenheim, Schafer and Buck, Discrete-Time Signal Processing, 2nd edition (Prentice-Hall, 1999)
- Mathematics of DSP
- References:
 - McClellan, Schafer, & Yoder, DSP First
 - Ifeachor Jervis “Digital Signal Processing–A Practical Approach”, Prentice Hall

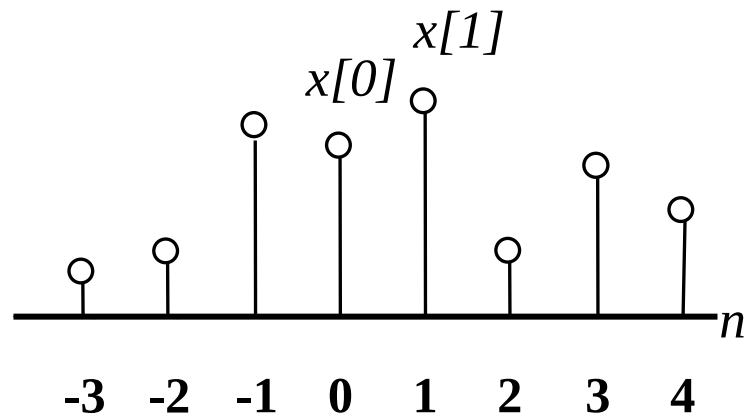
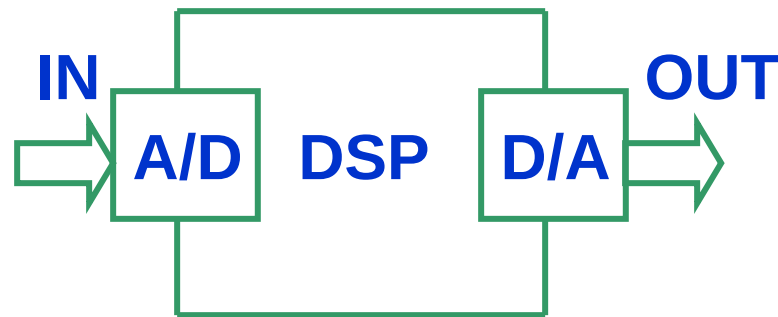
DSP Components

- converting analogue signals into a digital (usually binary) sequence
- performing all signal processing operations in the digital form
- if necessary, converting the digital information back to analogue signal



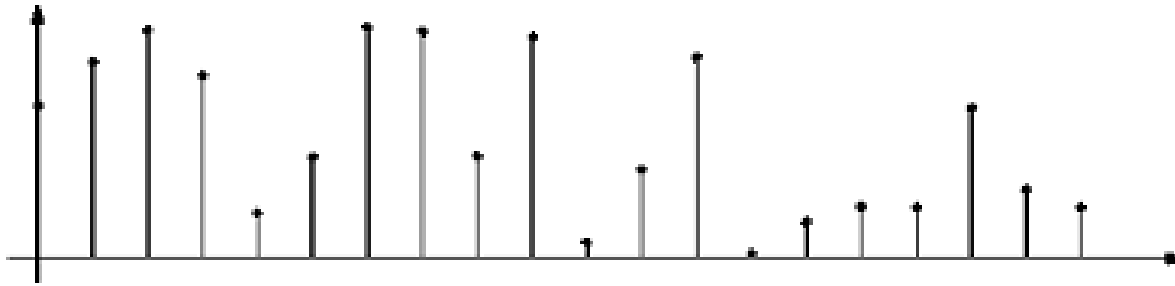
DSP Introduction

Application of mathematical operations
to digitally represented signals



Discrete Time Signal

sequence $x[n]$



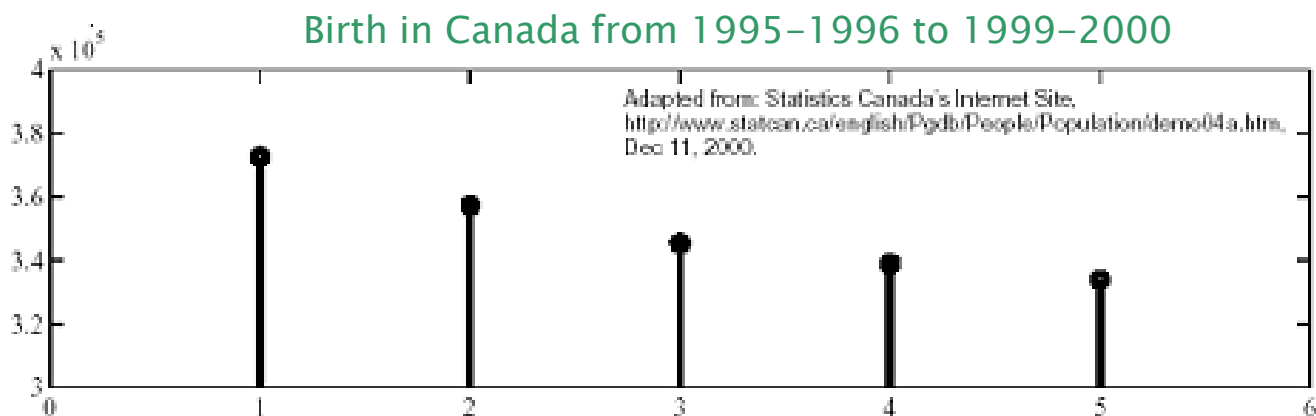
- as opposed to continuous-time signals $x(t)$
- “time” = independent variable

Discrete in Nature

- stock market indices

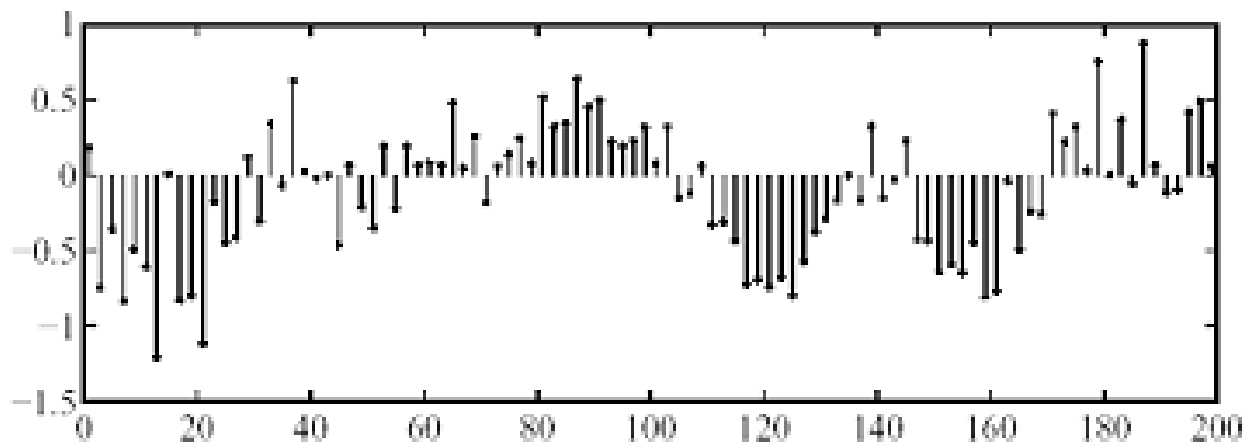
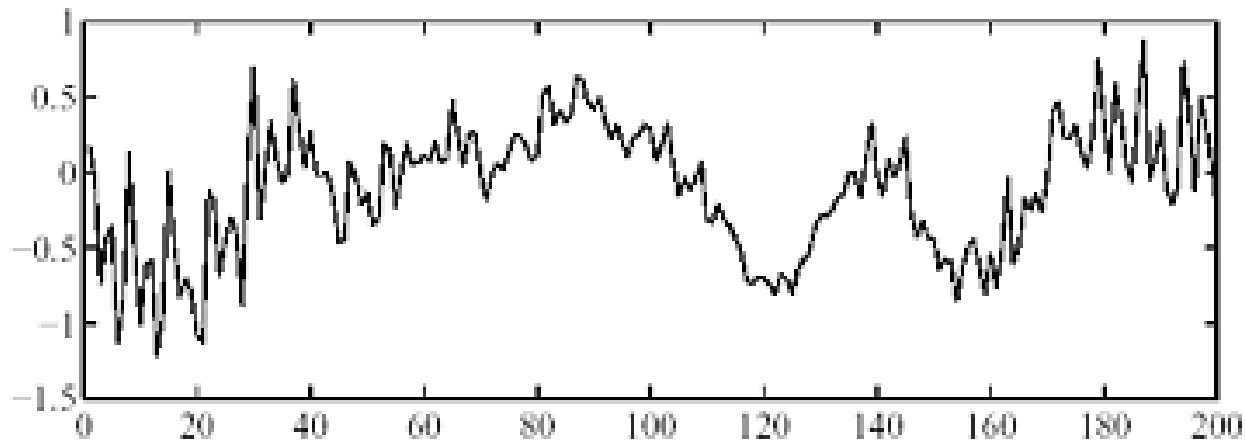


- population statistics



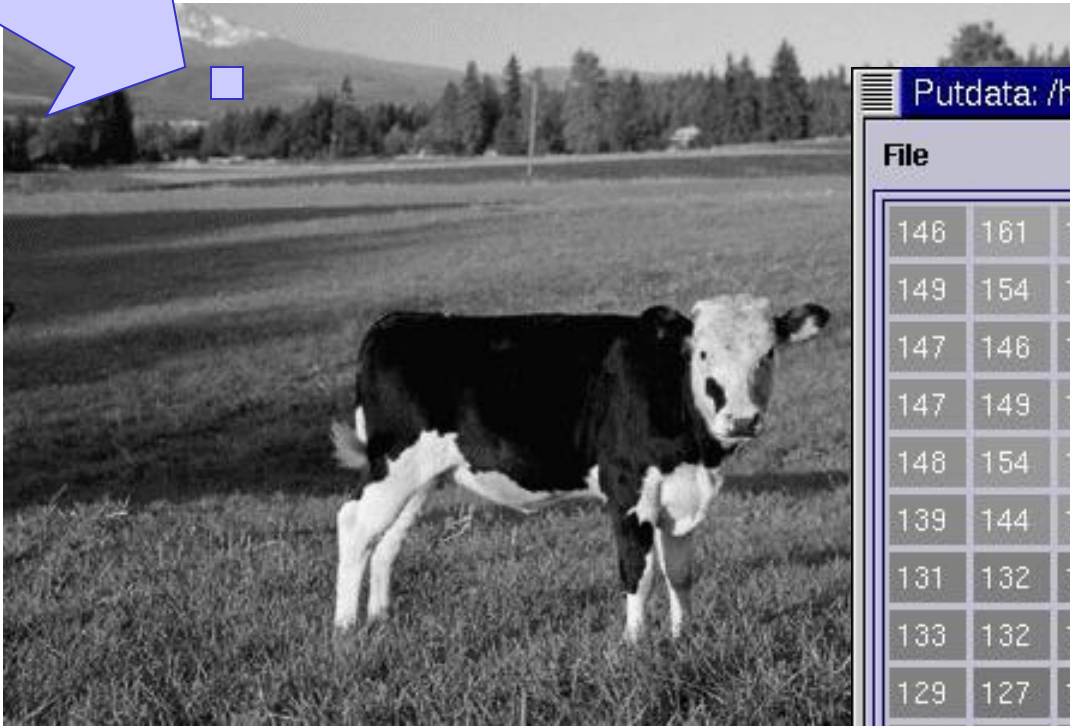
Example

Sampled continuous-time (analog) signals
– Speech



Digital Images

2-D arrays (matrices) of numbers



Putdata: /home/camps/cowgray.jpg									
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146	161	165	159	165	177	166	142	143	141
149	154	152	149	158	171	164	147	144	141
147	146	145	148	157	160	151	139	140	138
147	149	157	167	167	155	139	129	133	132
148	154	167	176	169	150	135	131	131	131
139	144	152	155	149	139	133	133	133	134
131	132	132	131	132	133	131	127	130	132
133	132	129	127	134	141	134	122	125	127
129	127	126	128	131	132	130	127	129	127
129	127	126	128	131	132	130	128	130	129